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Research article

Optimal demand response aggregation in wholesale electricity markets: Comparative analysis of polyhedral; ellipsoidal and box methods for modeling uncertainties

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ABSTRACT

This paper presents an optimal framework for demand response aggregators in wholesale electricity markets. Demand response aggregators provide customers with flexible contracts in the proposed model. These contracts allow for hourly load reductions through load curtailment, load shifting, onsite generation utilization, and energy storage systems. The study suggests the price-based self-scheduling model for demand response aggregators, which seeks to maximize the aggregator's payoff for participation in day-ahead energy markets. The proposed model is implemented on a sample demand response aggregator with uncertainty and without uncertainty. Three methods, box, polyhedral, and ellipsoidal, based on robust optimization, are proposed for uncertainty modeling in this work, and their effectiveness is compared with each other. The results show that if the uncertainty is modeled using the polyhedral method, the value of the objective function will deviate by only 22.72 % compared to the case without uncertainty. However, this deviation in box and ellipsoidal methods is 118.48 % and 49.20 %, respectively, which shows the superiority of the polyhedral method compared to the oval and box methods.

Nomenclature

A. Indices and abbreviations	
t	Index for the time.
z	Index of load reduction contracts.
LC	Load curtailment.
LS	Load shifting.
OG	Onsite generation.
ES	Energy storage.
B. Constants	
ρ_t	Electricity market price at hour t.
$R_{z,t}^{LS}$	Load reduction price in LS strategy.
$R_{z,t}^{OG}$	Load reduction price in OG strategy.

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(*continued*)

$R_{z,t}^{ES}$	Load reduction price in ES strategy.
$H_{z,t}^{LC}$ / $H_{z,t}^{LS}$	Aggregated load curtailment of registered customers in the LC and LS strategies.
$T_z^{ON,OG}$ / $T_z^{OFF,OG}$	The minimum duration of ON/OFF in each of the contracts in the OG strategy
$QD_z^{\max,LC}$ / $QD_z^{\min,LC}$	Maximum/Minimum duration of load reduction in each of the contracts in the LC strategy.
FS_z^{OG}	Bidding the startup cost in each of the contracts in the OG strategy.
SG_z^{OG}	Bidding the startup fuel in each of the contracts in the OG strategy.
RAP_z^{OG}	Ramp-up limit in each of the contracts in the OG strategy.
RAD_z^{OG}	Ramp-down limit in each of the contracts in the OG strategy.
$PWo_z^{max,ES}$	Power Rating in each of the contracts in the ES strategy.
$EC_z^{Max,ES}$	ES energy capacity in each of the contracts in the ES strategy.
MDN_z^{ES}	The maximum charge/discharge cycles of ES in each of the contracts in the ES strategy.
$KL_z^{\max,LC}$	Maximum number of daily load curtailments.
$POW_z^{\max,OG}$ / $POW_z^{\min,OG}$	Maximum/Minimum rate on power generation in each of the contracts in the OG strategy.
G_z^{OG}	Fuel coefficient in each of the contracts in the OG strategy.
$G_z^{Max,OG}$	The maximum amount of fuel consumption in each of the contracts in the OG strategy.
RAP_z^{ES} / RED_z^{ES}	Charging/Discharging ramp of ES in each of the contracts in the ES strategy.
${\eta_z}^{D,ES}$	Discharge efficiency of ES in each of the contracts in the ES strategy.
IC_z^{LC} / IC_z^{LS}	Load reduction initiation cost in each of the contracts in LC/LS strategies.
$ENRT_z^{Es}$	Energy retention time in each of the contracts in the ES strategy.
C. Variables	
$POW_{z,t}^{ES}$	ES Power Discharging in each of the contracts in the ES strategy at time t.
$POW_{z,t}^{OG}$	ES Power generation in each of the contracts in the ES strategy at time t.
$FGF_{z,t}^{OG}$	Startup fuel in each of the contracts in the ES strategy at time t.
$FG_{z,t}^{OG}$	Startup cost in each of the contracts in the ES strategy at time t.
Q_t^{LC}	Total scheduled load reduction in LC strategy at time t.
Q_t^{LS}	Total scheduled load reduction in LS strategy at time t.
Q_t^{OG}	Total scheduled load reduction in OG strategy at time t.
Q_t^{ES}	Total scheduled load reduction in ES strategy at time t.
$CQ_{z,t}^{LC}$	Total cost of load reduction in LC strategy.
$CQ_{z,t}^{LS}$	Total cost of load reduction in LS strategy.
$CQ_{z,t}^{OG}$	Total cost of load reduction in OG strategy.
$CQ_{z,t}^{ES}$	Total cost of load reduction in ES strategy.
$QIC_{z,t}^{LC}$	Load reduction initiation cost in each of the contracts in the LC strategy at time t.
$QIC_{z,t}^{LS}$	Load reduction initiation cost in each of the contracts in the LS strategy at time t.
$QIC_{z,t}^{OG}$	Load reduction initiation cost in each of the contracts in the OG strategy at time t.
$QIC_{z,t}^{ES}$	Load reduction initiation cost in each of the contracts in the ES strategy at time t.
$M_{z,t}^{LC}$	The state of load reduction in the LC strategy.
$M_{z,t}^{LS}$	The state of load reduction in the LS strategy.
$M_{z,t}^{OG}$	The state of load reduction in the OG strategy.
$F_{z,t}^{LC}$ / $S_{z,t}^{LC}$	The starting/stopping Load reduction initiation cost in each of the contracts in the LC strategy at time t.
$F_{z,t}^{LS}$ / $S_{z,t}^{LS}$	The starting/stopping load reduction initiation cost in each of the contracts in the LS strategy at time t.
$F_{z,t}^{OG}$ / $S_{z,t}^{OG}$	The starting/stopping load reduction initiation cost in each of the contracts in the OG strategy at time t.
$F_{z,t}^{ES}$ / $S_{z,t}^{ES}$	The starting/stopping Load reduction initiation cost in each of the contracts in the ES strategy at time t.
Ω	Ellipsoidal set uncertainty variable.
ω	Polyhedral set uncertainty variable.
ψ	Box set uncertainty variable.
ϴ	Auxiliary variable for polyhedral and ellipsoidal modeling.

1. Introduction

With the emergence of technologies such as smart metering infrastructure and smart electrical devices, the need for demand-side management is felt more and more [1]. China introduced the concept of demand-side management in 1990, but the idea was officially published in 2010 b y the National Development and Reform Commission [2]. Based on studies conducted in 2018, Beijing, Suzhou, Tangshan, and Foshan have been comprehensively investigated for electricity demand management for the first time in the world [3]. Demand response can match supply and demand in a flexible manner [4,5]. The US department of Energy divides demand response methods into two categories: in the first category, the time of use of equipment is changed according to time-based tariffs to minimize the price of electricity [6]. Therefore, such consumers are suitable for intraday or day-ahead consumption adjustments [7]. In the second category, customers reduce their energy consumption by participating in incentive programs [8]. Consumers can earn through incentive schemes [9]. Therefore, participation on the demand side is known as an optimal and economical solution for electricity markets, as well as reducing peak demand, creating a “solid balance” in the electricity industry, and increasing the reliability of electrical energy systems [10,11]. Demand side management helps power companies in critical power grid conditions by balancing supply and demand [12,13]. Demand response has many advantages, including adapting to the environment [14]. Also, demand

Table 1
Summarize of related work.

Ref	Uncertainty modeling method	Load reduction strategies				Focus	self-scheduling model	Customer-Centric
		ESS	LC	LS	OG			
[18]	Robust optimization	✓	×	×	×	Supply side	×	×
[19]	Scenario-based	×	✓	×	×	Demand side	×	×
[20]	Without uncertainty	✓	✓	✓	✓	Supply& demand side	✓	✓
[21]	Robust optimization	×	×	✓	×	Demand side	×	×
[22]	IGDT	✓	×	×	×	Supply side	×	×
[23]	Hybrid stochastic-Robust	✓	×	×	×	Supply side	×	×
[24]	Scenario-based	✓	×	×	×	Supply side	×	×
[25]	Without uncertainty	×	✓	✓	×	Demand side	×	×
[26]	Without uncertainty	✓	×	✓	×	Supply& demand side	×	×
[27]	Without uncertainty	✓	×	×	✓	Supply side	×	×
[28]	Without uncertainty	✓	×	×	✓	Supply side	×	×
This work	Polyhedral; Ellipsoidal &Box methods based- Robust optimization	✓	✓	✓	✓	Supply& demand side	✓	✓

response can reduce fossil fuel consumption, and creating stability and optimal management will help to integrate renewable energy into the power grid. In general, six basic applications have been defined for smart grids: advanced metering infrastructure, demand response management, distributed generation and storage resources, distribution automation, and power transmission. Also, using these technologies promises to increase demand response in the operation of the electricity market. On the other hand, energy regulatory bodies update electricity market rules and remove barriers to demand response integration in electricity markets. For example, under the Federal Energy Regulatory Commission's Order 719, ISOs in the United States were forced to accept demand response offers compared to other sources in wholesale markets [15]. Although demand response will be able to provide system services in wholesale electricity markets, it will be provided in distribution systems to industrial, residential, and commercial consumers. Now, with the aim of optimal use of demand response programs, a new concept called demand response aggregator has been introduced [16]. Demand response aggregators can aggregate price-based resources to participate in demand response programs [17].

In reference [18], a demand response aggregator is responsible for participating in the wholesale electricity market. In this research, the uncertainty of the price of electricity is considered, and the robust optimization method is used to model the uncertainty. Reference [19], proposes the participation of demand response aggregators in wholesale electricity marketers that manage contracts to participate in day-ahead and real-time markets. In this study, the uncertainty of the electricity price is modeled using the scenario-based method on historical data. Also, in Ref. [20], demand response aggregators propose different types of flexible contracts to load reduction on consumers in wholesale electricity markets. In Ref. [21], in order to change the consumption patterns of customers, a competitive model between demand response aggregators to conclude a demand response service contract with the network operator is presented. Reference [22], proposes a new structure using commercial demand response aggregators for the participation of the commercial sector in the electricity market. In this study, electricity price uncertainty is modeled by the Information-Gap Decision Theory (IGDT) method. In Ref. [23], the objective is to increase the demand response aggregator profit. In this work, the authors have classified different types of customers. In Ref. [24], an optimization model for the participation of a distributed energy resource aggregator in the day-ahead market in the presence of demand flexibility is presented. In another research, the aim is to load reduction and the aggregator signs LS and LC contracts with its customers [25]. In Ref. [26], the optimal bidding strategy of the demand response aggregator is discussed based on the modeling of customer response behaviors under different incentives. Reference [27], proposes a new decision framework for demand response aggregator business in wholesale and retail markets. In Ref. [28], an optimization framework for optimal participation of an aggregator in the day-ahead market through direct control of a set of ES is presented.

Related works provide valuable insights but have some research gaps that will be addressed below:

- Wholesale market prices fluctuate due to various factors, such as market psychology, supply chain disruptions, and demand fluctuations. Some related works that do not consider uncertainty may provide misleading insights into market dynamics and future trends. There is also an essential gap in electricity market research regarding developing new methods because the modeling methods presented in related works try to depict the complexities of this dynamic environment, which leads to incorrect predictions and suboptimal decision-making for market participants.
- The price-based self-scheduling model empowers consumers to react to real-time price signals. When prices are high, they can choose to reduce consumption, shifting demand to lower-cost periods. This helps balance supply and demand in real-time, reducing the need for expensive peaking power plants. Research that doesn't account for price-based models wouldn't explore the most effective strategies for consumers to schedule their energy use based on dynamic pricing.
- Customers are classified into some related work, which creates challenges. Customers with unique consumption patterns may be placed in a class that does not reflect their actual needs. This can discourage them from participating in load reduction contracts

Therefore, according to the mentioned research gaps, the authors in this paper express their contributions as follows:

- This paper presents a new approach for modeling wholesale market price uncertainty using robust optimization. Our method uses a flexible framework with box, polyhedral, and ellipsoidal uncertainty sets. This framework accommodates different risk profiles, and due to the fact that it does not need a probability distribution function in its operational process, it can be tractable as quickly as possible. Therefore, it offers a significant advantage compared to traditional uncertainty modeling methods.
- This work proposes an optimization framework for demand response aggregators in day-ahead energy markets. The proposed framework provides a customer-centric approach that increases flexibility and potential participation in demand response programs. Demand response aggregators utilize the proposed price-based self-scheduling model to determine its optimal participation schedule in day-ahead energy markets. Also, demand response aggregators have used a price-based self-scheduling model to optimize their participation in day-ahead markets, creating a win-win situation for themselves, consumers, and the electricity grid.

Table 1 summarizes the related work and innovations related to this work.

This paper is organized as follows:

Section 2, explains the concepts under discussion in this study. The objective function of this work, relationships related to load reduction contracts and robust optimization model are presented in Section 3. In addition, in section 4, of this research, the introduction of the input data has been discussed. In Section 5, the simulation results are discussed. Finally, Section 6, deals with the conclusion and future works.

2. Concepts under discussion

2.1. Demand response aggregator model

Traditionally, electricity grids relied solely on power plants to meet fluctuating demand. Demand response aggregators bring consumers into the equation. Incentivizing consumers to adjust their electricity usage during peak hours effectively reduces demand when the grid is stressed. Therefore, by balancing supply and demand fluctuations, they enable a more reliable and responsive power grid. Fig. 1 illustrates how an Independent System Operator (ISO) initiates demand response programs. The ISO transmits information to aggregators in the day-ahead market, prompting them to submit bids on behalf of their customers. Acting as intermediaries, these aggregators actively engage with the ISO and consumers to maximize program participation. This hierarchical framework ensures all customers can participate in demand response programs. It's important to note that aggregators can be existing market participants like distribution system operators or load-serving entities, depending on the market structure. This paper will consider demand response aggregators as financially independent actors separate from system operators. These aggregators are supposed to operate by providing customer demand response programs to wholesale electricity markets.

2.2. Contract related to demand response

Demand response aggregators can use different programs, such as capacity and energy programs, as well as different categories of customers and load reduction strategies [29]. This section treats the LC provided by demand response customers as a negative load. As seen in Fig. 2, four types of contracts are classified for different classes of customers in the day-ahead energy market for demand response aggregators. Typically, demand response aggregators focus on specific courses of load reduction methods rather than covering all demand response options. However, in this paper, all the ways of responding to the possible demand for the future market have been thoroughly investigated, which will be explained below.

2.3. Robust optimization

In mathematical programming, almost all existing problems are solved by assuming the data is definite in advance. But in reality, the available information is uncertain [30]. However, in mathematical programming, model development is based on known data, in which data uncertainty has no impact on the quality and feasibility of the answers [31]. As a result, due to the change of one of the existing data, many of the applied restrictions may be violated and lead to non-optimization of the obtained answer [32]. As a result, the main goal is to provide a method that is resistant to this data uncertainty and provides an optimal response [33]. In recent years, robust optimization has been widely studied from the view of uncertainty management [34]. In general, a robust optimization approach will be able to provide a solution with two main features as follows [35]:

- Evaluation of the practicality of the proposed method is given credit for any unknown parameter realization. At the same time, there is no need for a large volume of scenarios, and only the minimum and maximum amount of the parameter is required.
- Robust optimization finds the optimal solution for all possibilities of the uncertain [36,37].

This paper applies robust optimization for uncertainty modeling for two reasons. First, obtaining the distribution function for a variable with uncertainty is difficult. Secondly, an index will not be added to the problem in the robust optimization method [38].

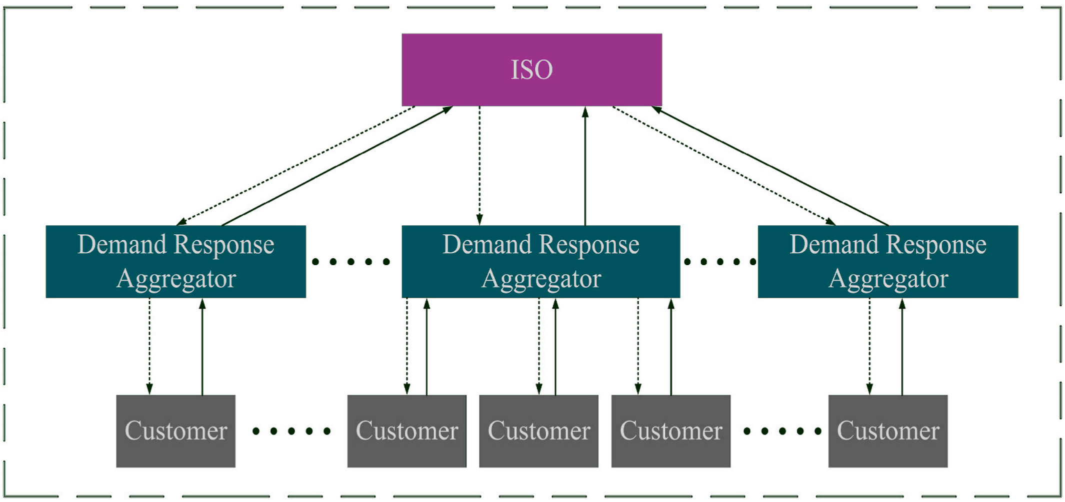

Fig. 1. Structure and operation of demand response aggregators.

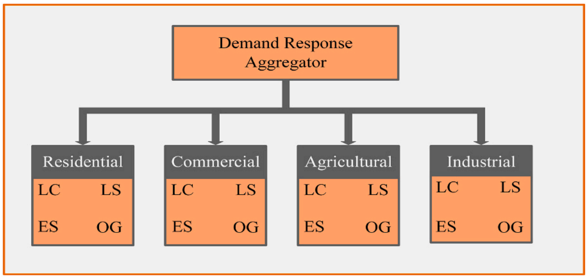

Fig. 2. Classification of demand response contracts in future energy markets for different customer classes.

However, the robust optimization method has no concern about this issue [39]. Fig. 3 shows the integration of box, polyhedral and ellipsoidal methods in the optimization model.

Fig. 4 shows the degree of conservatism of the three box, polyhedral, and ellipsoidal methods. ψ shows the level of conservatism of the box method, which models the worst cases and has a high conservatism compared to the elliptic and polyhedral methods. Then Ω and ω, which show the conservatism of ellipsoid and polyhedral, respectively, are more conservative. Therefore, under uncertainty, the polyhedral method provides the closest solution to the optimal solution.

3. Problem formulations

3.1. Investigating a proposed self-scheduling model for demand response aggregators

In this paper, the main objective of the aggregator is to maximize his return (i.e., total revenue minus variable costs) in the day-ahead market. In addition, aggregators profits are derived from the sale of demand response products to electricity markets and from the reduction of hourly electricity prices for customers in demand response programs. Therefore, the price difference in the contracts of customers participating in the demand response programs and the hourly cost of the energy market will have a positive result for the aggregators. Typically, demand response aggregators focus on specific classes of load reduction methods rather than covering all demand response options. However, in this paper, all the ways of responding to the possible demand of the future market have been thoroughly investigated, which will be explained further. As shown in equation (1), the objective function consists of two parts, the

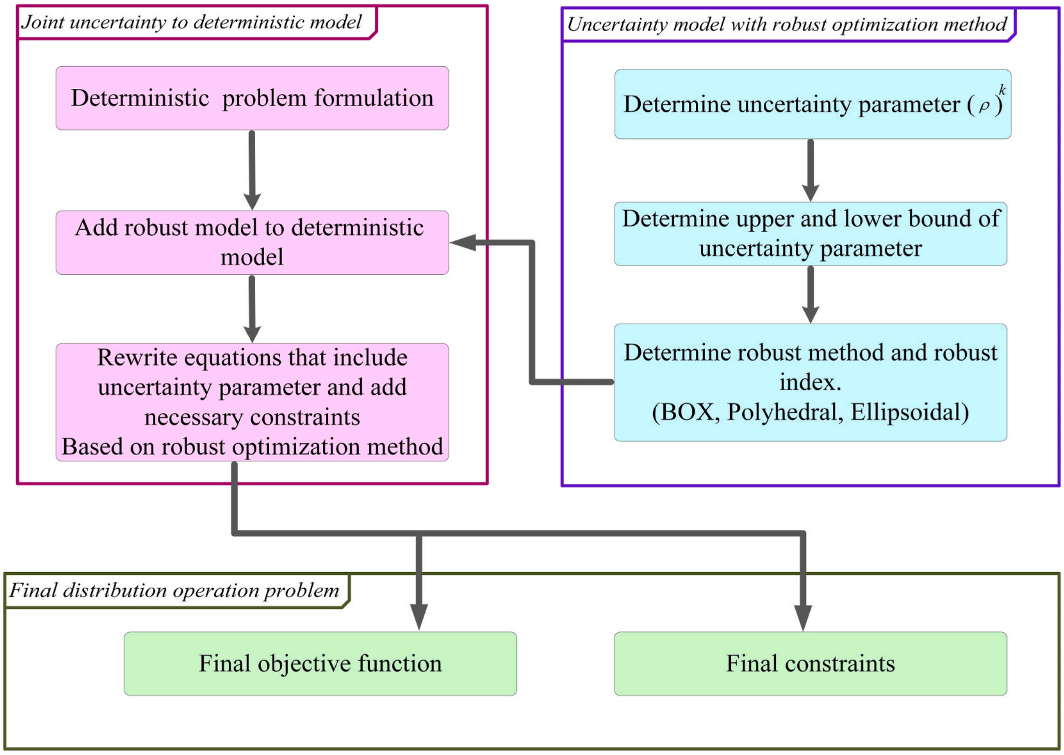

Fig. 3. Model-based framework for robust optimization.

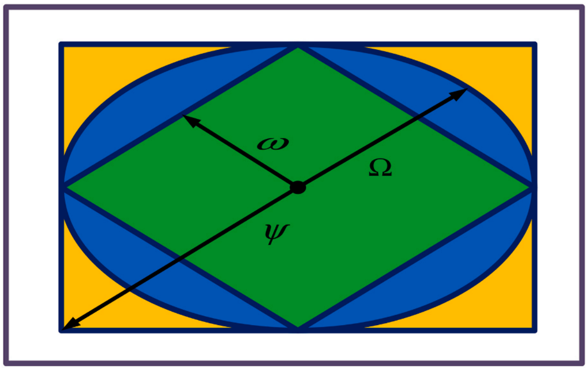

Fig. 4. Comparing the optimization results of three ellipsoidal, box and Polyhedral models in the presence of uncertainty.

first part is related to the income from the sale of LS, ES, and OG strategies. In the objective function, aggregators compensate for load reduction by more accurately predicting energy market prices, which is usually possible using numerical techniques such as time series and neural networks [20,40]. The second part of the objective function is related to the costs paid to the customers who participated in load reduction programs. Also, aggregators can predict hourly electricity market prices by analyzing historical electricity market price data. Therefore, the method proposed in this paper is based on the results of multiple demand response aggregators. Other market factors on market prices are considered in the self-scheduling price prediction. Other market factors that influence market prices are considered in the prediction of self-scheduling prices. The equations related to load reduction methods in this paper will be effective on the objective function mentioned in the following sections.

$$Maximize \sum_{t \in NT} \left[\rho_t \left(Q_t^{LC} + Q_t^{LS} + Q_t^{OG} + Q_t^{ES} \right) - \left(CQ_t^{LC} + CQ_t^{LS} + CQ_t^{OG} + CQ_t^{ES} \right) \right] \tag{1}$$

3.2. LC constraints

LC strategy describes the old and traditional methods to reduce the load. For example, in this contract, industrial consumers may lessen some of their unnecessary activities, or household consumers should limit their consumption for the period the demand responders have specified a reduction in the contract [41]. The costs of load reduction $R_{z,t}^{LC}$ result from the agreement between the customers and the aggregators. And also the total amount $H_{z,t}^{LC}$ of reducing the load of the customers in each of the LC strategy contracts. It should also be mentioned that the evaluations of the aggregators ability to respond to the demand of the customers determine the amount of load reduction. The following equation shows the minimum and maximum period of load reduction, the maximum amount of load reduction, and the start costs of load reduction in LC contracts. According to equations (2) and (3), the binary variable $M_{z,t}^{LC}$ changes from 0 to 1 when programmed by demand response aggregators. Equation (4) shows the costs related to the start of load shedding. The load reduction in each contract is represented by a binary variable that is 1 if it starts at time t and 0 otherwise. Equations (5) and (6) show the minimum and maximum period of load reduction, equation (7) the maximum amount of load reduction daily, and equation (8) the contract start/stop. Finally, equation (9) represents a binary variable in that only one item can be 1 at any moment.

$$Q_t^{LC} = \sum_{Z \in N_{LC}} H_{z,t}^{LC} M_{z,t}^{LC} \tag{2}$$

$$CQ_t^{LC} = \sum_{z \in N_{LC}} \left(QIC_{z,t}^{LC} + R_{z,t}^{LC} + H_{z,t}^{LC} + M_{z,t}^{LC} \right) \tag{3}$$

$$QIC_{z,t}^{LC} \geq IC_z^{LC} F_{z,t}^{LC} \quad \forall z, \forall t \tag{4}$$

$$\sum_{t'=t}^{t+qD_Z^{\min,LC}-1} M_{z,t'}^{LC} \geq QD_z^{\min,LC} F_{z,t}^{LC} \quad \forall z, \forall t \tag{5}$$

$$\sum_{t'=t}^{t+QD_Z^{\max,LC}-1} STI_{z,t'}^{LC} \geq SRI_{z,t}^{LC} \quad \forall z, \forall t \tag{6}$$

$$\sum_{t \in T} F_{z,t}^{LC} \leq KL_z^{\max,LC} \quad \forall z \tag{7}$$

$$F_{z,t}^{LC} - S_{z,t}^{LC} = M_{z,t}^{LC} - M_{z,(t-1)}^{LC} \quad \forall z, \forall t \tag{8}$$

$$F_{z,t}^{LC} + S_{z,t}^{LC} \leq 1 \quad \forall z, \forall t \tag{9}$$

3.3. LS constraints

In this strategy, energy consumption is not limited to any period until the total load demand specified in the contract is met in a given period [41]. For example, industrial or residential consumers can postpone using equipment with high energy consumption until a few days late [42]. Equations (10) and (12) show the cost function, the cumulative load reduction, and the start cost of load reduction in this contract. Also, equations (13) and (14) show the minimum and maximum duration of load reduction. As can be seen in equations (15) and (16), three sets of binary variables $M_{z,t}^{LS}$, $F_{z,t}^{LS}$, $S_{z,t}^{LS}$ are defined to indicate the start and stop of the contract at time t. The equations between binary variables are shown in equations (15) and (16). In this contract, some data points prove that customers in the future energy market will change their load value from period T_z^{LS} to period T_z^{SH}. Equation (17) shows the guarantee for aggregators that the LS contract will be delivered in a period T_z^{SH} .Also, relevant information and data will be delivered to ISO by the aggregator according to the schedule that is transferred in this contract.

$$Q_t^{LS} = \sum_{z \in N_{LS}} H_{z,t}^{LS} M_{z,t}^{LS} \tag{10}$$

$$CQ_t^{LS} = \sum_{z \in N_{LS}} \left(QIC_{z,t}^{LS} + R_{z,t}^{LS} + H_{z,t}^{LS} + F_{z,t}^{LS} \right) \tag{11}$$

$$QIC_{z,t}^{LS} \geq IC_z^{LS} F_{z,t}^{LS} \quad \forall z, \forall t \tag{12}$$

$$\sum_{t'=t}^{t+QD_z^{\min,LS}-1} M_{z,t'}^{LS} \geq QD_z^{\min,LS} F_{z,t}^{LS} \quad \forall z, \forall t \tag{13}$$

$$\sum_{t'=t}^{t+QD_z^{\max,LS}-1} S_{z,t'}^{LS} \geq F_{z,t}^{LS} \quad \forall z, \forall t \tag{14}$$

$$F_{z,t}^{LS} - S_{z,t}^{LS} = M_{z,t}^{LS} - M_{z,(t-1)}^{LS} \quad \forall z, \forall t \tag{15}$$

$$F_{z,t}^{LS} + S_{z,t}^{LS} \leq 1 \quad \forall z, \forall t \tag{16}$$

$$S_{z,t}^{LS} = 0 \quad \forall t \notin T_z^{LS} \tag{17}$$

3.4. OG constraint

In this strategy, consumers traditionally support generation operations by replacing a backup energy source [43,44].

$$Q_t^{OG} = \sum_{z \in N_{OG}} POW_{z,t}^{OG} \tag{18}$$

$$CQ_t^{OG} = \sum_{z \in N_{OG}} \left(FG_{z,t}^{OG} + V_{z,t}^{OG} X_{z,t}^{OG} \right) \tag{19}$$

$$FG_{z,t}^{OG} \geq FS_z^{OG} \left(M_{z,t}^{OG} - M_{z,(t-1)}^{OG} \right) \quad \forall z, \forall t \tag{20}$$

$$Z_{z,t}^{OG} POW_z^{\min,OG} \leq POW_{z,t}^{OG} \leq M_{z,t}^{OG} POW_z^{\max,OG} \quad \forall z, \forall t \tag{21}$$

$$POW_{z,t}^{OG} - POW_{z,(t-1)}^{OG} \leq RAP_z^{OG} \quad \forall z, \forall t \tag{22}$$

$$POW_{z,(t-1)}^{OG} - POW_{z,t}^{OG} \leq RAD_z^{OG} \quad \forall z, \forall t \tag{23}$$

$$\sum_{t'=t}^{t+T_z^{N,OG}-1} M_{z,t'}^{OG} \geq T_z^{ON,OG} \left(M_{z,t}^{OG} - M_{z,(t-1)}^{OG} \right) \quad \forall z, \forall t \tag{24}$$

$$\sum_{t'=t}^{t+T_z^{F,OG}-1}\left(1-M_{z,t'}^{OG}\right) \geq T_z^{Off,OG}\left(M_{z,(t-1)}^{OG}-M_{z,t}^{OG}\right) \quad \forall z, \forall t \tag{25}$$

$$\sum_{t \in T}\left(G_z^{OG}POW_{z,t}^{OG}+FGF_{m,t}^{OG}\right) \leq G_z^{Max,OG} \quad \forall z \tag{26}$$

$$FGF_{z,t}^{OG} \geq SG_z^{OG}\left(M_{z,t}^{OG}-M_{z,(t-1)}^{OG}\right) \quad \forall z, \forall t \tag{27}$$

3.5. ES system constraints

With the proposal of "peak system carbon goals and neutrality" the advent of ES is rapidly increasing in power [45,46]. Also, these systems can be classified into five parts based on ES, which are thermal, mechanical, chemical, electrochemical, and electrical. Large electricity customers can use ESs in different ways [47,48]. This paper focuses on electric ES systems that may consist of connections to several small distribution storage units. In this strategy, customers generally provide part of their energy needs from their ES system. Equations (28) and (29) show the total load reduction supplied by demand response aggregators through ES contracts at time t. Equation (30) indicates that the total load reduction is lower than the ES power's. Also, equations (31) and (32) show the discharge rate of ES. Equation (33) guarantees the aggregator that the total ES capacity in the contract in the planned period, considering the discharge efficiency, is not greater than the ES generation source. Also, the minimum amount of charging and discharging is shown in equation (34). Equation (35) shows the amount of remaining energy in the zth contract. Equations (36) and (37) represent the binary variables related to this contract.

$$Q_t^{ES} = \sum_{z \in N_{ES}} POW_{z,t}^{ES} \tag{28}$$

$$QIC_t^{ES} = \sum_{z \in N_{ES}} R_{z,t}^{ES}POW_{z,t}^{ES} \tag{29}$$

$$0 \leq POW_{z,t}^{ES} \leq M_{z,t}^{ENST}PWo_z^{\max,ES} \quad \forall z, \forall t \tag{30}$$

$$POW_{Z,t}^{ES}-RAD_{Z,(t-1)}^{ES} \leq RAP_z^{ES} \quad \forall z, \forall t \tag{31}$$

$$POW_{z,(t-1)}^{ES}-POW_{z,t}^{ES} \leq RED_z^{ES} \quad \forall Z, \forall t \tag{32}$$

$$\sum_{t \in T} POW_{z,t}^{ES} \leq {\eta_z}^{D,ES}EC_z^{Max,ES} \quad \forall z \tag{33}$$

$$\sum_{t \in T} F_{z,t}^{ES} \leq MDN_z^{ES} \quad \forall z \tag{34}$$

$$\sum_{t'=t}^{t+ENRT_z^{Es}-1} S_{z,t'}^{ES} \geq SRI_{z,t}^{ES} \quad \forall z, \forall t \tag{35}$$

$$F_{z,t}^{ES}-S_{,t}^{ES} = M_{z,t}^{ES}-M_{z,(t-1)}^{ES} \quad \forall z, \forall t \tag{36}$$

$$F_{z,t}^{ES}+S_{z,t}^{ES} \leq 1 \quad \forall z, \forall t \tag{37}$$

in this paper, equations (1)–(37) show the self-scheduling model aggregating demand response for the future without considering uncertainty. Demand response aggregators charge ES systems and restore shed load in LS contracts based on hourly forecasts when energy prices are low. However, in this case, it could lead to an unexpected load increase in a few hours, which complicates the operation of the ISO and may compromise the system's security. In this research, the optimal participation of a demand response aggregator in the next-day energy market has been used to determine the market price, and this optimal participation will be determined according to the hourly price of energy and the amount of contract load reduction.

3.6. Robust optimization constraints

We use equation (38) between a ρ max and min to show the uncertainty variable. Then, equations (39) and (40) respectively indicate the difference between (average) and a (perturbation).

$$\rho_t^{\min} \leq \rho_t \leq \rho_t^{\max} \; \forall t \tag{38}$$

$$\overline{\rho_t} = \frac{1.1\rho_t + 0.9\rho_t}{2} \quad \forall t \tag{39}$$

$$\widehat{\rho_t} = \frac{1.1\rho_t - 0.9\rho_t}{2} \quad \forall t \tag{40}$$

Different robust optimization methods offer varying degrees of robustness. Some methods prioritize finding the absolute best solution under the worst-case scenario (highly robust), while others aim for a balance between optimality and robustness. Equation (41) shows the box uncertainty set introduced by Soyster [49]. It is a linear model that prioritizes robustness over optimality and ensures feasibility even under extreme parameter changes. The box uncertainty set may be too conservative for some problems. On the other hand, the conservatism of the oval uncertainty set is lower compared to the box uncertainty set. Equations (42) and (43) show the modeling related to ellipsoidal uncertainty set. This model is a convex non-linear programming problem. Bertsimas and Sim present a new model of the uncertainty set to improve the previous methods, called a polyhedral uncertainty set [50,51]. This model has the lowest level of conservatism compared to the box and ellipsoidal uncertainty set and leads to a linear model. Equations (44) and (45) introduce the polyhedral uncertainty set.

$$Z \le \sum_t \left[\left(\overline{\rho_t} - \psi \widehat{\rho_t} \right) \left(CQ_t^{LC} + CQ_t^{LS} + CQ_t^{OG} + CQ_t^{ES} \right) \right] \tag{41}$$

$$Z \le \sum_t \left(\overline{\rho_t} Q_t^{LC} + Q_t^{LS} + Q_t^{OG} + Q_t^{ES} \right) - (\Omega)(\theta) \tag{42}$$

$$\theta^2 \ge \sum_t \left[\widehat{\rho_t} Q_t^{LC} + Q_t^{LS} + Q_t^{OG} + Q_t^{ES} \right]^2 \tag{43}$$

$$Z \le \sum_t \left(\overline{\rho_t} Q_t^{LC} + Q_t^{LS} + Q_t^{OG} + Q_t^{ES} \right) - (w)(\theta) \tag{44}$$

$$\theta \ge \widehat{\rho_t} Q_t^{LC} + Q_t^{LS} + Q_t^{OG} + Q_t^{ES} \tag{45}$$

4. Input data

Table 2 shows the data related to three LC contracts. Also, the data related to the LS contract are presented in Table 3. Finally, the data of OG and ES contracts are shown in Tables 4 and 5, respectively. Also, the hourly price during the day is shown in Fig. 5, which can be seen according to this figure that the peak load occurred at 17:00.

5. Simulation results

In this work, two case studies are considered:

- Case study 1: Without uncertainty
- Case study 2: With uncertainty

In case study 1, uncertainty is not considered in the optimization model, and the results show a load reduction of customers by participating in LC, LS, OG, and ES contracts. By considering uncertainty, stakeholders can make informed decisions, manage risk, and ensure the smooth operation of the electricity market. Therefore, in case study 2, the uncertainty of the wholesale market price is considered. Three box, polyhedral, and ellipsoidal methods based on robust optimization have been used to model the desired uncertainty parameter. The results obtained in this case study compare the amount of customers' load reduction in each of the strategies provided by the demand response aggregators in the three proposed methods for uncertainty modeling.

5.1. Simulation results in case study 1

The prices related to the electricity market in this research work are based on the final costs of PJM, which were presented in 2012.

Table 2
Statistical data related to LC contracts.

Contract number	Measure (MW)	Price ($/MW)	Load reduction initiation expensive ($)	Min/max Load reduction during (h)	Highest number of daily load reduction
1	10	40	100	(3,6)	1
2	10	45	100	(3,6)	1
3	10	50	100	(3,6)	1

Table 3
Data related to LS contracts.

Contract number	T_z^{LS}	T_z^{SH}	α
1	10–16	4–10	%100
2	10–14	8–14	%100
3	16–22	10–16	%100

Table 4
Statistical data related to OG contracts.

Contract number	Minimum power (MW)	Maximum power (MW)	Price ($/MW)	Initiation expense	Minimum on and off time (h)	Ramp up and down	Fuel initiation expense (MBTU)	Extent fuel (MBTU)
1	1	10	40	100	1	10	20	100
2	1	10	45	100	1	10	20	100
3	1	10	50	100	1	10	20	100

Table 5
Statistical data related to ES contracts.

Contract number	Power rank (MW)	Price ($/MW)	Energy capacity (MWh)	Discharge efficiency	Discharge Ramp (MW/h)	Energy preservation time (h)
1	10	40	60	0.9	20	12
2	10	45	60	0.9	20	12
3	10	50	60	0.9	20	12

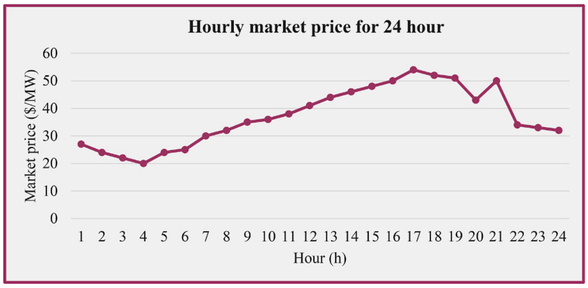

Fig. 5. Hourly price curve during the day.

As seen in Fig. 6, the LS and LC output curves at 0:00 to 15 and 19–24 h, the hourly schedules are 0 if the market price is less than 35 ($/MWh). However, in the curve related to OG, it can be seen that the customers participate in relatively more hours than in LC and LS mode. And even when the market price is only 5 ($/MWh), few customers experience it. According to the output curve, the number of customers decreases in the ES state compared to all three states, OG, LS, and LC. Between 0:00 and 16:00 and 19:00 to 24:00, the participation rate is zero, and the customers only participate in the programs for 4 h. However, as can be seen in the comparison of the 5 output curves, ES is the only mode that, unlike the previous three curves of the customers between 16:30 and 17:30, participates in the highest price in the market, as seen in the LS and LC output curves at 0:00–15 and 19–24 h, if the market price is less than 35 ($/MWh), customers will not participate in demand response programs. The only difference between the curves of LS and LC is in the volume of their participation. LS participates in demand response programs about 5 MW more than LC. However, in the curve related to OG, it can be seen that the customers participate in relatively more hours than in LC and LS mode. And even when the market price is only 5 ($/MWh), few customers experience it. According to the output curve, the number of customers in the ES mode is reduced compared to all three OG, LS, and LC methods. The participation rate is zero from 0:00 to 16:00 and from 19:00 to 24:00, and customers only participate in programs for 4 h. However, as can be seen in the comparison of the 4 output curves, ES is the only mode in which, unlike the previous three curves, customers participate in demand response programs between 16:30 and 17:30 in a high volume equal to 45 MW.

5.2. Simulation results in case study 2

From the viewpoint of the demand response aggregator in the wholesale electricity market, the reduction of electricity market prices leads to a decrease in the profit of the demand response aggregator. Furthermore, the worst case of electricity market prices is

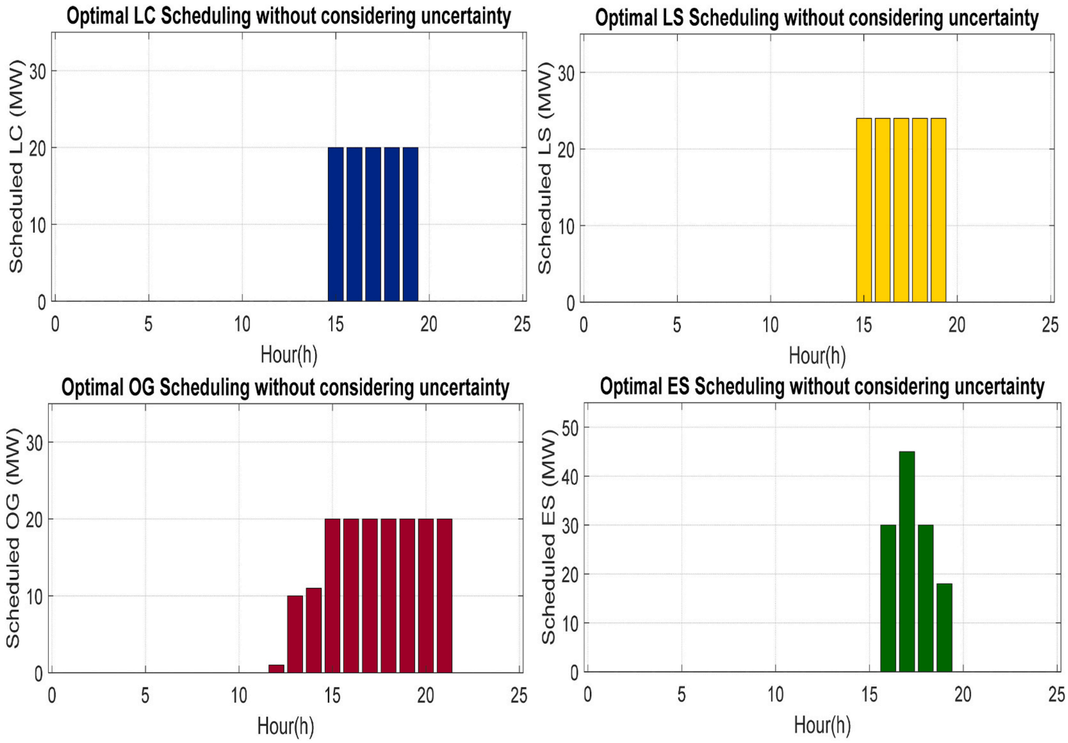

Fig. 6. Comparing the results of modeling four load scheduling strategies.

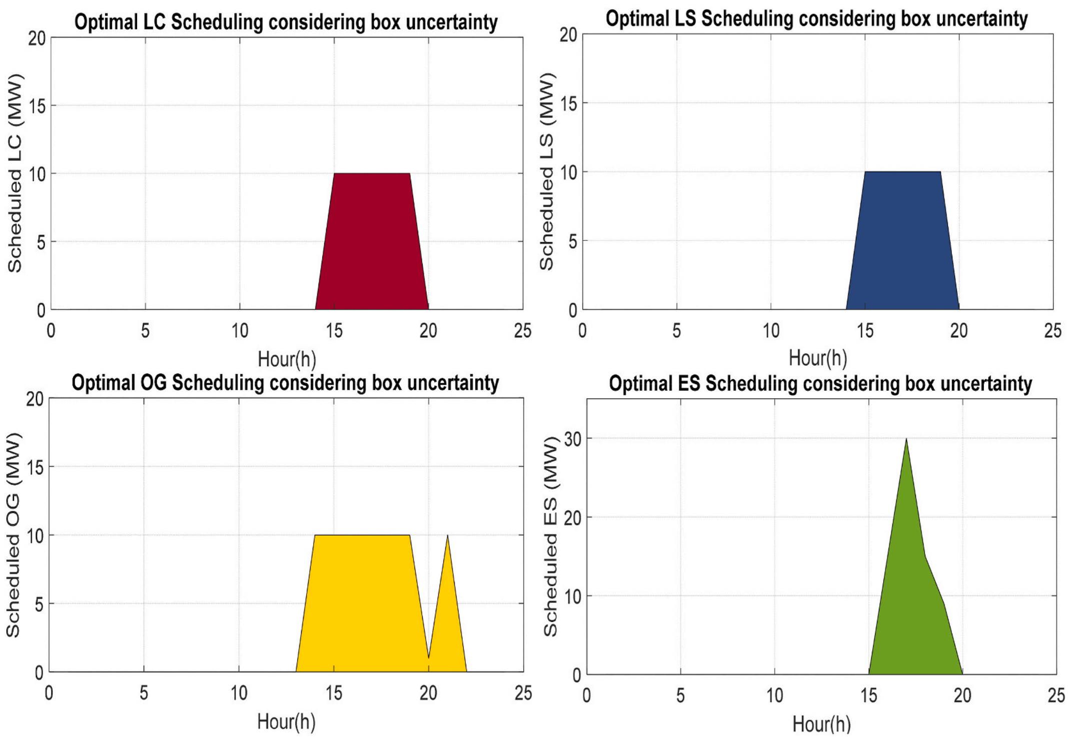

Fig. 7. Result related to the box model under uncertainty.

considered in the uncertainty modeling via three-box, polyhedral, and ellipsoidal methods based on a robust optimization approach. Therefore, the profit of the aggregator will be reduced in cases considering uncertainty. Fig. 7 shows the modeling of the box method under uncertainty. In this method, customers in LC and LS contracts participate in demand response programs for about 5 h. Also, in this method, the customers are less inclined to reduce their load than in the case without uncertainty, and they reduce their load about 50–60 % less than in the case without uncertainty, which shows that the box method models the worst-case. In the OG contract, some customers participate in demand response programs for about 3 h, reducing their load by a minimal amount. Some customers then participate in demand response programs for about 6 h. Finally, in the ES contract, the number of customers participating in demand response programs is much less than in the earlier contracts. Also, on the other hand, the load reduction of customers in this contract is about 18 % more than that of the LC, OG, and LS contracts.

Fig. 8 shows the modeling of the ellipsoidal method under uncertainty. According to the figure, the total amount of customer load reduction in the ellipse method is more than in the box method, and uncertainty modeling in the ellipsoidal method will be more optimal than in the box method. The box method is conservative, leading to less load reduction due to its focus on worst-case scenarios. The ellipsoidal method is a more realistic approach, allowing for more significant load reduction by considering the probability distribution of prices.

In addition, the participation of consumers in flexible contracts in the conditions of modeling uncertainties by the polyhedral method is shown in Fig. 9. According to the figure, the load reduction in the uncertainty modeling method using the polyhedral method is greater than that of the box and ellipsoidal method. In this case, the results are close to the situation where the uncertainty in the electricity price is not taken into account. This issue can be related to their appearance. A polyhedral is a 3D shape with flat faces. This allows them to create more intricate and specific regions to represent uncertainty in electricity prices. This flexibility enables the polyhedral method to capture the nuances of price behavior, potentially leading to more targeted strategies for load reduction. Also, the smoothness of ellipsoids makes them a powerful tool for depicting the general landscape of price uncertainty, which is crucial for informed decision-making in various market settings. Based on the analysis and comparison of Figs. 8 and 9, it can be seen that this method is less accurate compared to the polyhedral method, but it is still effective in load reduction. The box method is the least complicated among these three methods. Therefore, as previously shown in Fig. 7, it overestimates the potential price volatility, leading to overly conservative strategies for load reduction. This could mean less flexibility compared to ellipsoidal and polyhedral methods.

Fig. 10, shows the sensitivity analysis based on the box, polyhedral, and ellipsoidal methods to compare the performance of these three methods better. Therefore, the robust index is moving step by step from 0 to 1 to show that as it approaches 1, the value of the objective function is also reduced and considering that the objective is maximization, we are moving away from the optimal value.

Table 6, presents the numerical results related to case studies 1 and 2. The goal of the proposed self-scheduled model is to maximize the payoff (that is, the difference between the total revenue and cost) in the day-ahead market. Therefore, the objective function value is the same as the income. To further investigate the deviations, a statistical graph is used to provide a visual representation of the results, as seen in Fig. 11.

6. Conclusion

This paper presented an optimal framework for demand response aggregators, based on which demand response aggregators offer customers LC, LS, OG, and ES contracts to reduce the hourly load. Also, this paper presents an energy price-based self-scheduling model for a demand response aggregator to maximize the aggregator's efficiency in participating in future energy markets. In this work, the results with and without considering uncertainty have been analyzed as two case studies. Box, ellipsoidal, and polyhedral are our three proposed methods for modeling the uncertainty of wholesale market price. According to the sensitivity analysis performed on these three proposed methods, it can be concluded that the box method is highly conservative compared to the polyhedral and ellipsoidal methods. When limited information about the uncertainty is available, assuming the worst possible case can be helpful because the system is prepared to deal with it. Therefore, the box method may be the optimal choice in this situation. However, when a better understanding of the uncertainties is available, the polyhedral and ellipsoidal methods can lead to better optimization results.

Finally, in the modeling of the proposed method, the following results can be obtained:

1. Among the box, ellipsoidal, and polyhedral methods for uncertainty modeling, the polyhedral method exhibits the slightest deviation in the objective function's value compared to case study 1 (without uncertainty).
2. Load shedding flexibility offered through ES contracts compared to other contracts
3. The LS contract is more applicable than the LC contract, and the reason is that there is no load limit.
4. Reduction of data sent to ISO by demand response aggregators.
5. Maximizing the profits of aggregators.

In the end, the objective function of the problem was modeled by considering the uncertainty, and the result was that the polyhedral method has a more optimal value than the box and ellipsoidal methods. Also, researchers can use some of the works in this field in future research, some of which are as follows:

- Considering loads as uncertainty and using the column optimization method.
- Using the data-driven optimization method to model the uncertainties of the problem.
- Considering the extensive energy resources in the problem stated in this paper.

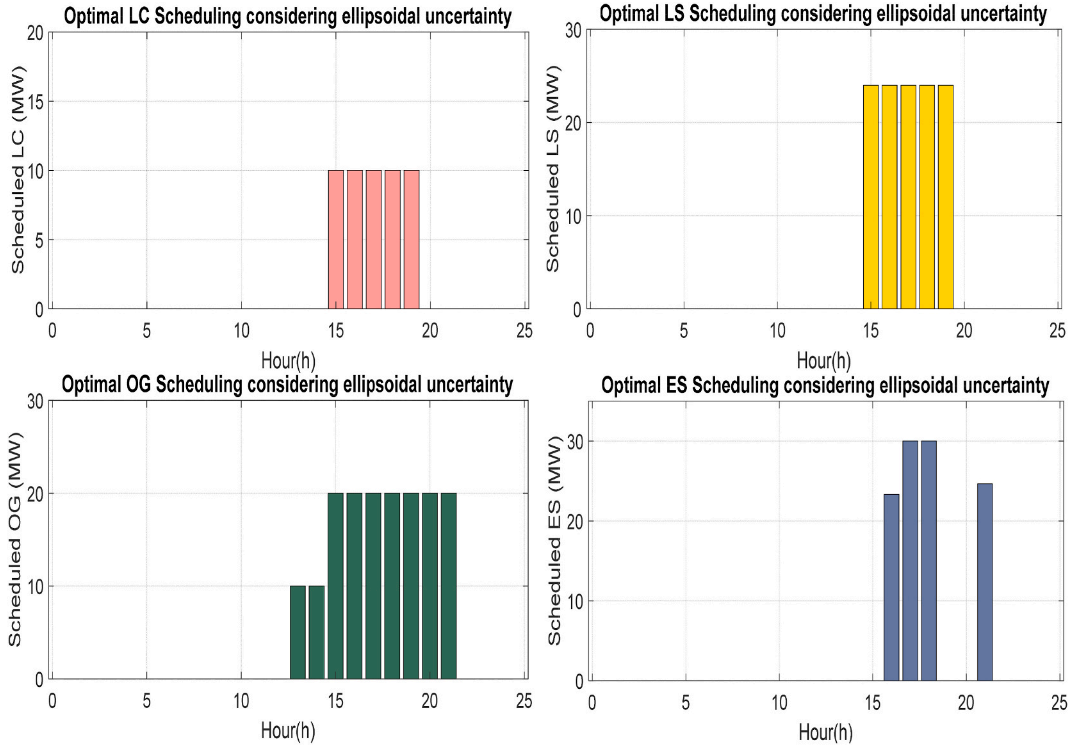

Fig. 8. The results of the ellipsoidal model under uncertainty.

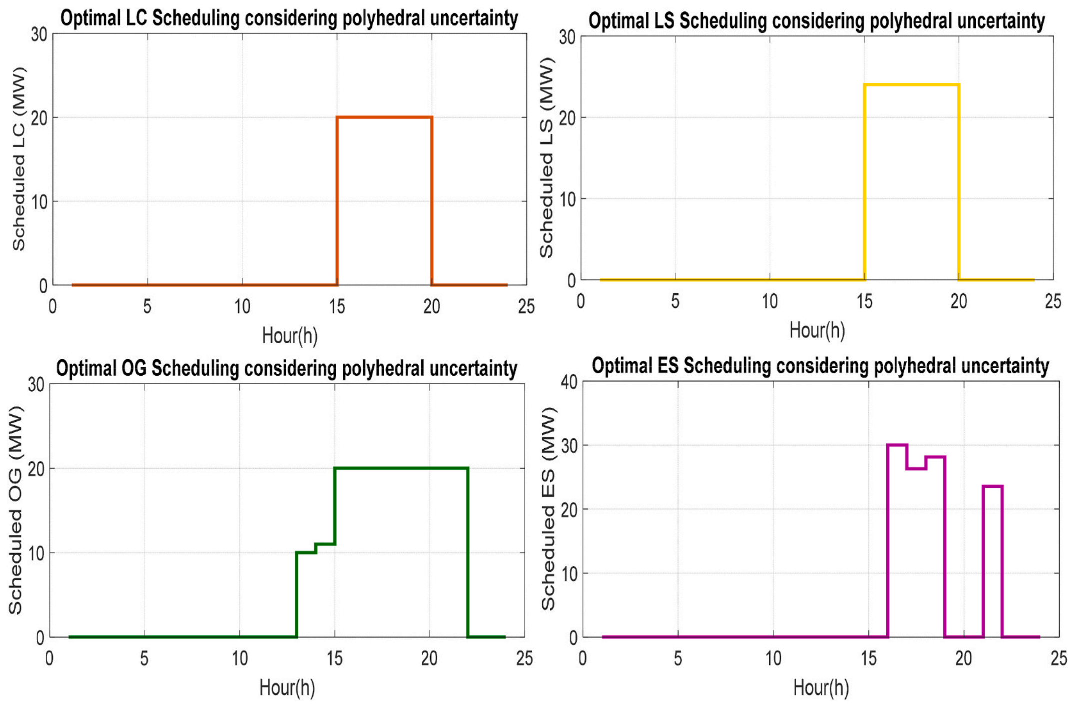

Fig. 9. Shows the results of the polyhedral method under uncertainty.

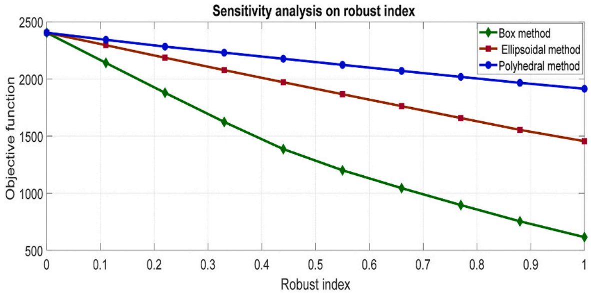

Fig. 10. Analysis of the sensitivity resulting from the modeling of three methods (Box, Polyhedral, Ellipsoidal).

Table 6
Numerical results obtained from simulation in conditions without uncertainty and uncertainty.

Objective function (Income)	Case study 1	Case study 2		
		Box model	Ellipsoidal model	Polyhedral model
Numerical result ($)	2404	615.3	1454.7	1913.6

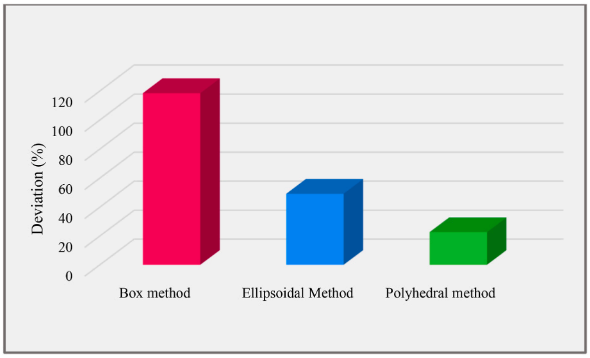

Fig. 11. The amount of deviation of the objective function in case study 2 compared to case study 1.

- Multi-targeting the objective function, such as adding the objective of reducing losses and using a multi-objective robust optimization method to model uncertainties.
- Using machine learning-based methods to model uncertainty.
- Modeling the uncertainty of the load demand side and the wholesale market price using the Column-and-Constraint Generation method.

CRediT authorship contribution statement

Nojavan Sayyad: Writing – review & editing, Writing – original draft, Supervision, Project administration, Conceptualization. **Mehrdad Tarafdar Hagh:** Writing – review & editing, Writing – original draft, Visualization, Validation. **Kamran Taghizad-Tavana:** Writing – review & editing, Writing – original draft, Software, Formal analysis, Data curation. **Mohsen Ghanbari-Ghalehjoughi:** Writing – review & editing, Writing – original draft, Resources, Methodology, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

None.

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